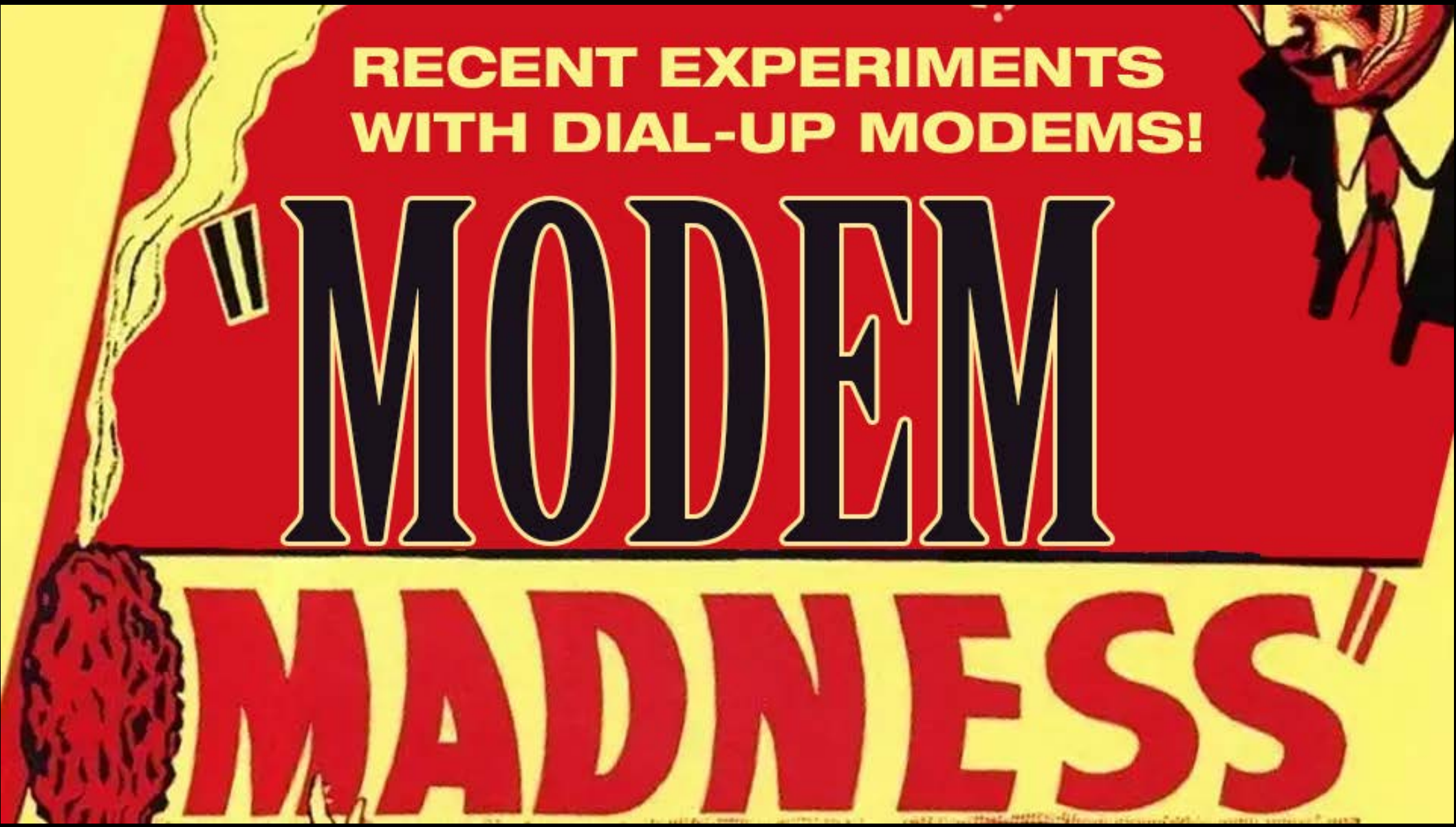

RECENT EXPERIMENTS
WITH DIAL-UP MODEMS!
"MODEM
MADNESS"

Agenda

- Who are we?
- Previously with Mike and Naveen
- Modems? Dial-up?
- Dialup World
- Other modem offerings?
  - NAPLPS-Server, Leitch Clock System Driver, Telecommunications Devices for the Deaf (TDDs)
- State of dial-up today
  - Dial-up war stories, multilink PPP
- Future Efforts

Who are You?

Mike

- Software engineer by trade, more excited when he can use a soldering iron
- Has been doing weird VoIP stuff for 13 years
- Roleplays as a telephone company
- Notices every manhole cover he walks past

Naveen

- Computer engineer with IT background
- Longtime phone phreak
- Scans payphones using modems
- Created PhreakNet
- Asterisk/DAHDI/C developer
- BBS sysop
- Environmental activist

Previously with Mike and Naveen...

- **PhilTel** is a telephone collective aiming to install free-to-use payphones in the Philadelphia area
- Mike created the “Protel Programming Line” so people with *Protel* payphones can freely call this service to make their payphone work again
- Naveen runs NIPCAQ, a system that perpetually calls hundreds of COCOTs for health and usage statistics



What is a Modem?

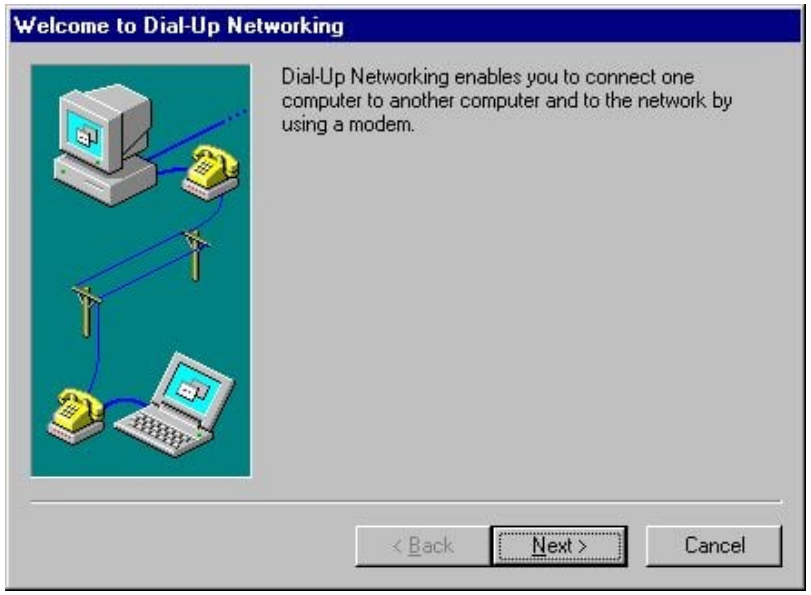
- **Modulator/Demodulator**
- A computer hardware device that converts data from a digital format into a format suitable for an analog transmission medium such as telephone or radio
- They come in many forms

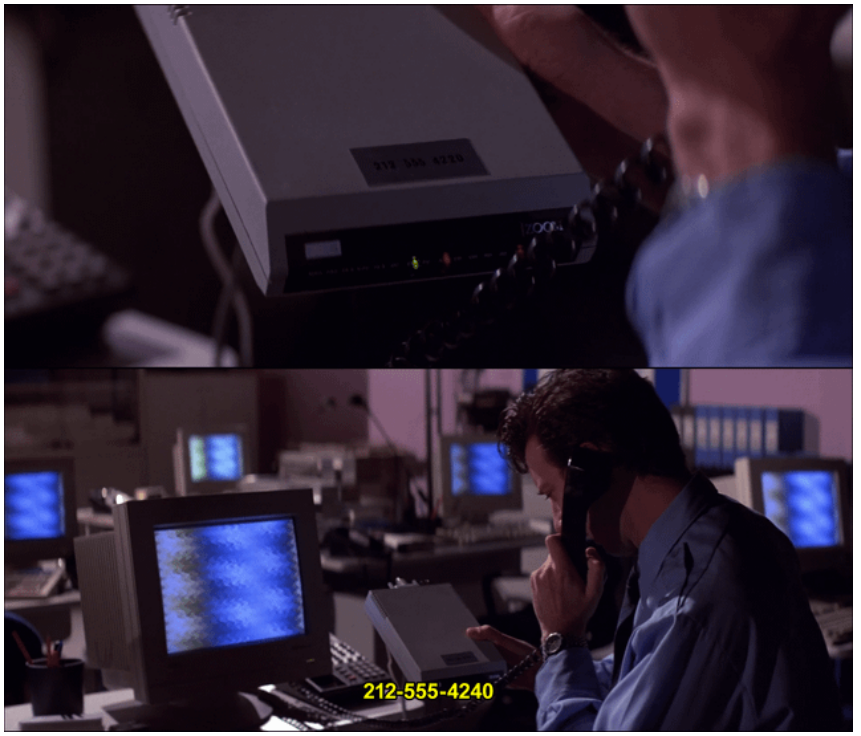




What is a Dial-up?

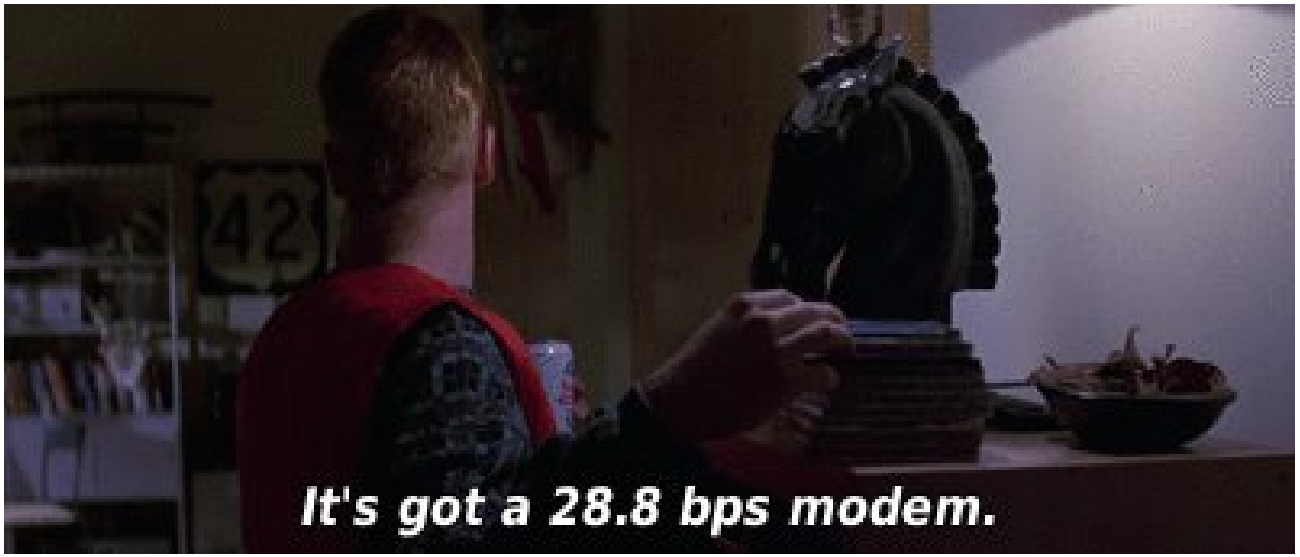
- Transmit computer data over an ordinary switched telephone line that has not been designed for data use
- Dial-up is often associated with Internet access, but can describe any way of accessing hardware/systems this way
- In the ‘90s, millions of Americans had dial-up Internet access, utilized by 75% of households
- Largely surpassed by broadband





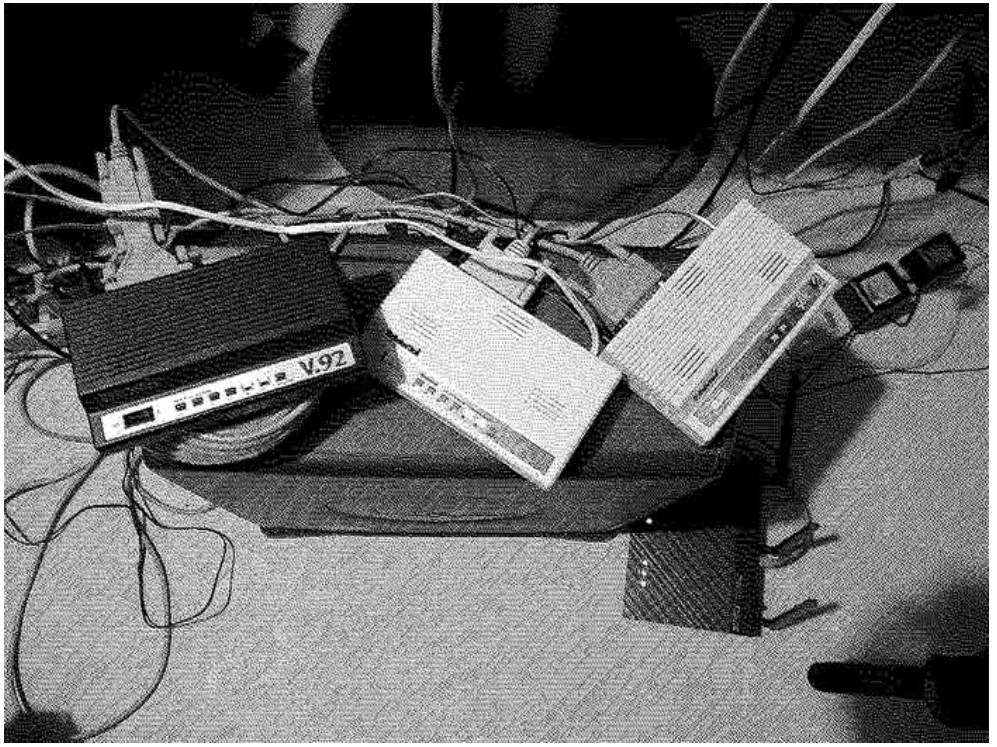
What is a Dial-up?

- Speeds are measured in bps (bits per second)
  - You are probably used to speeds in Mbps or Gbps
- Different speeds are usually associated with different standards
  - Bell Standards, Bell 101 = 110 bps, Bell 103 = 300 bps, Bell 212A = 1200 bps
  - ITU-T Standards, V.34 = 33.6K, V.90/V.92 = 56K



Dialup World

- A pretty bad (but functioning) dial-up ISP – http://dialup.world
  - Established 2021
  - Essentially built on trash
  - First iteration leveraged Server 2003 and Remote Access Service



Dialup World

- Currently runs on a Le Potato SBC with mgetty/pppd on Armbian
- Based on the guide from Doge Microsystems - https://dogemicrosystems.ca/wiki/Dial_up_server
- You can call it! **484-820-1337**
- 33.6K speeds! (Limitation with analog-analog communication)
- 4x Sportster modems
- Easter eggs



Dialup World

- WebTV Compatible
  - Connects to WebTV Redialed, an effort to reverse-engineer the original WebTV service - https://webtv.zone/
  - Works automagically with your old WebTV





Dialup World



- How does it connect to the PSTN?
  - VoIP number through Voip.ms
  - Grandstream ATAs to an Asterisk switch
- You can use a VoIP provider and ATAs for dial-up?!?
  - Yes, it works!
  - Force ulaw (G.711) codec
  - Set fixed jitter buffer, low
  - Turn off echo cancellation

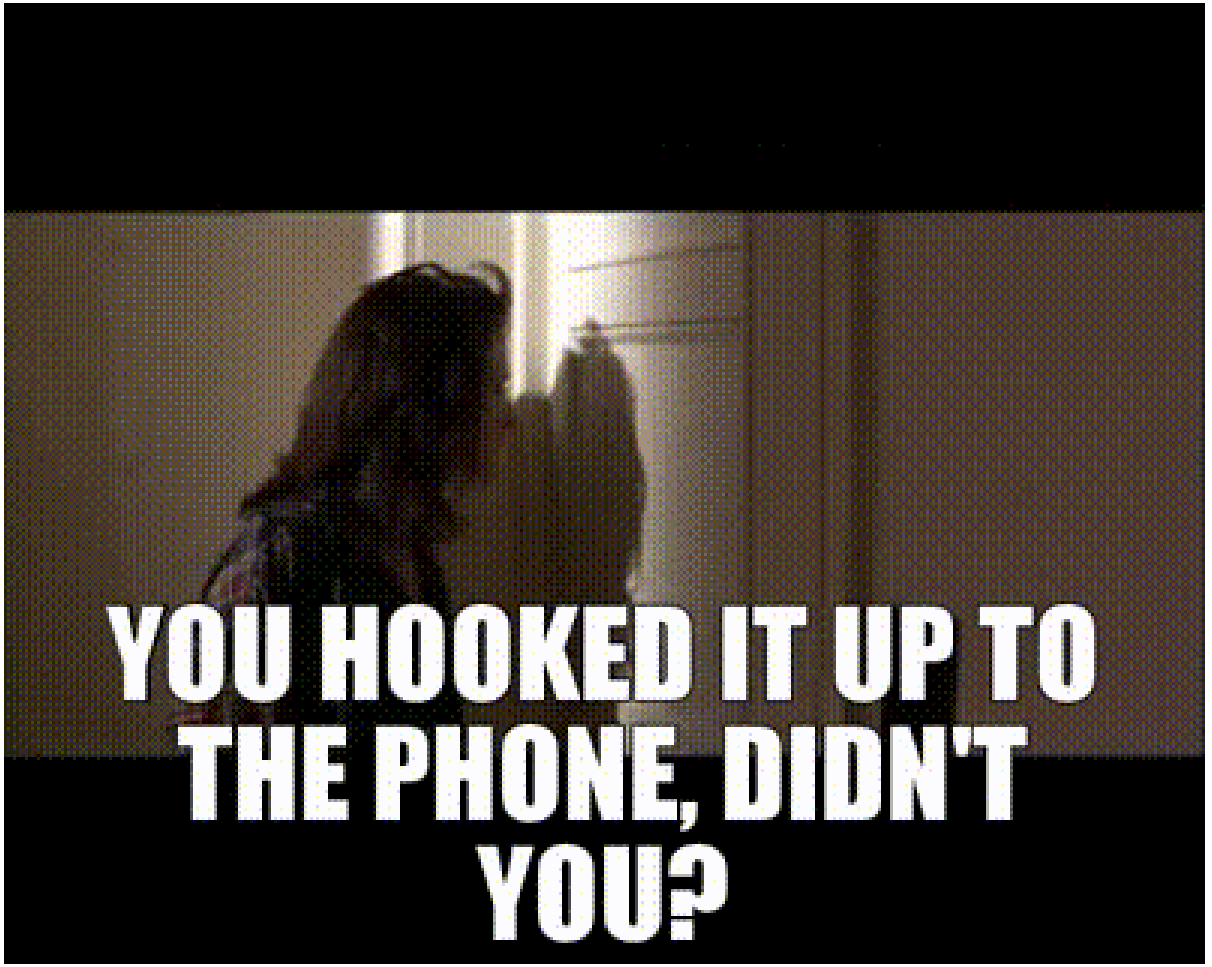


Dialup World

- There's renewed interest in Dial-up and vintage enterprise networking
  - **@theserialport**, **@clabretro** on YouTube
  - DosLab - https://dialin.doslabelectronics.com/
  - 2600.network - https://2600.network/



What else can we run?



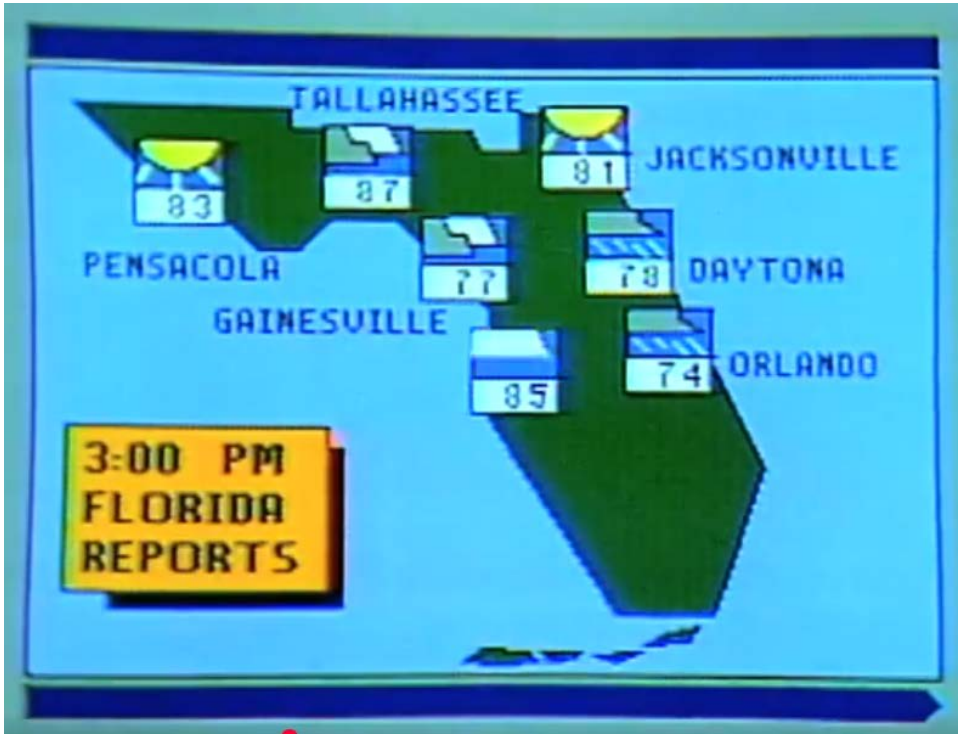
AT&T Sceptre

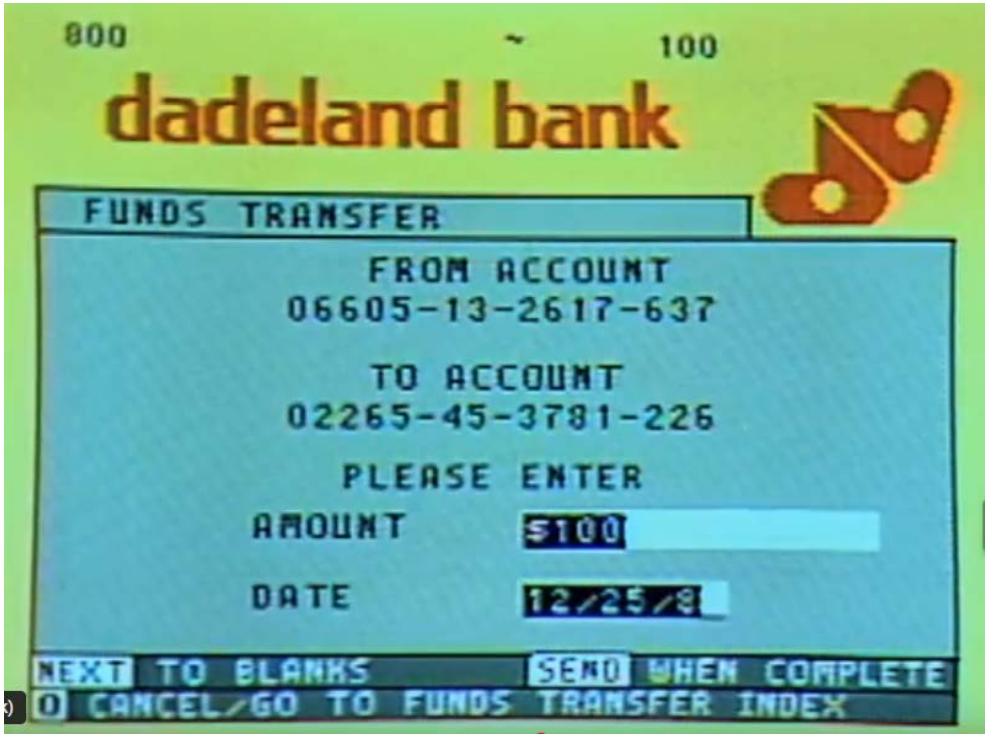
- An 80's videotex terminal that supports NAPLPS (North American Presentation Layer Protocol Syntax), used for ~3 years
- Originated from the Canadian Telidon Service



NAPLPS

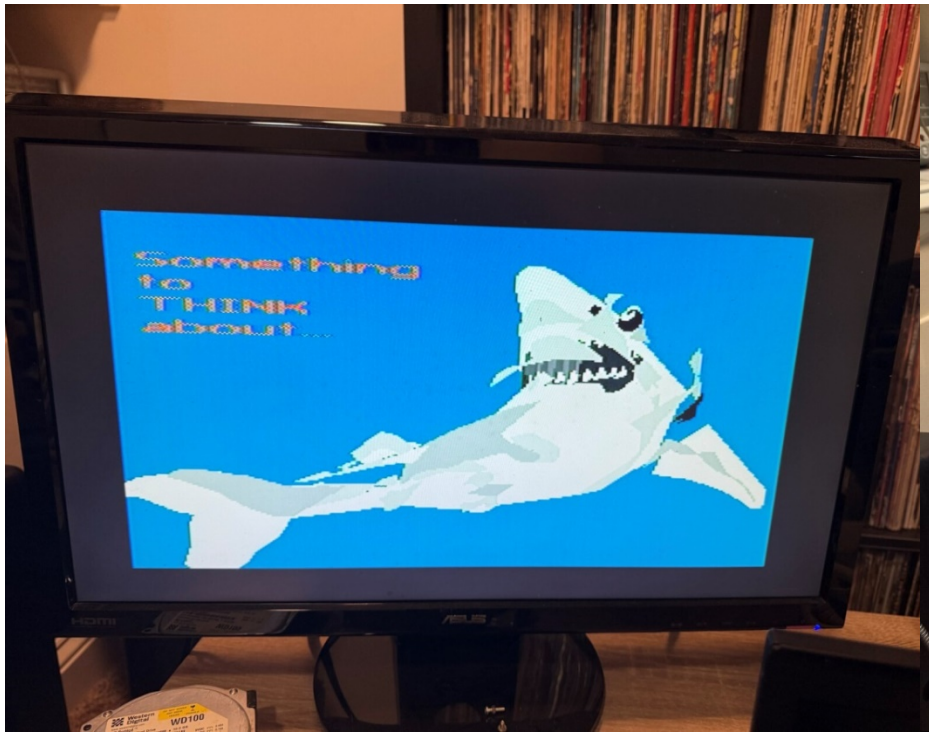
- A vector graphics format, used often as a GUI
- NAPLPS services often had a lot of interactivity such as online banking, flight booking, weather reports, and news





NAPLPS-Server

- I wrote a python application that interfaces with a modem to answer incoming calls and send NAPLPS file contents byte-by-byte
- The code is OSS - https://github.com/dialup-world/naplps-server
- There's a live demo number: **267-921-1337**





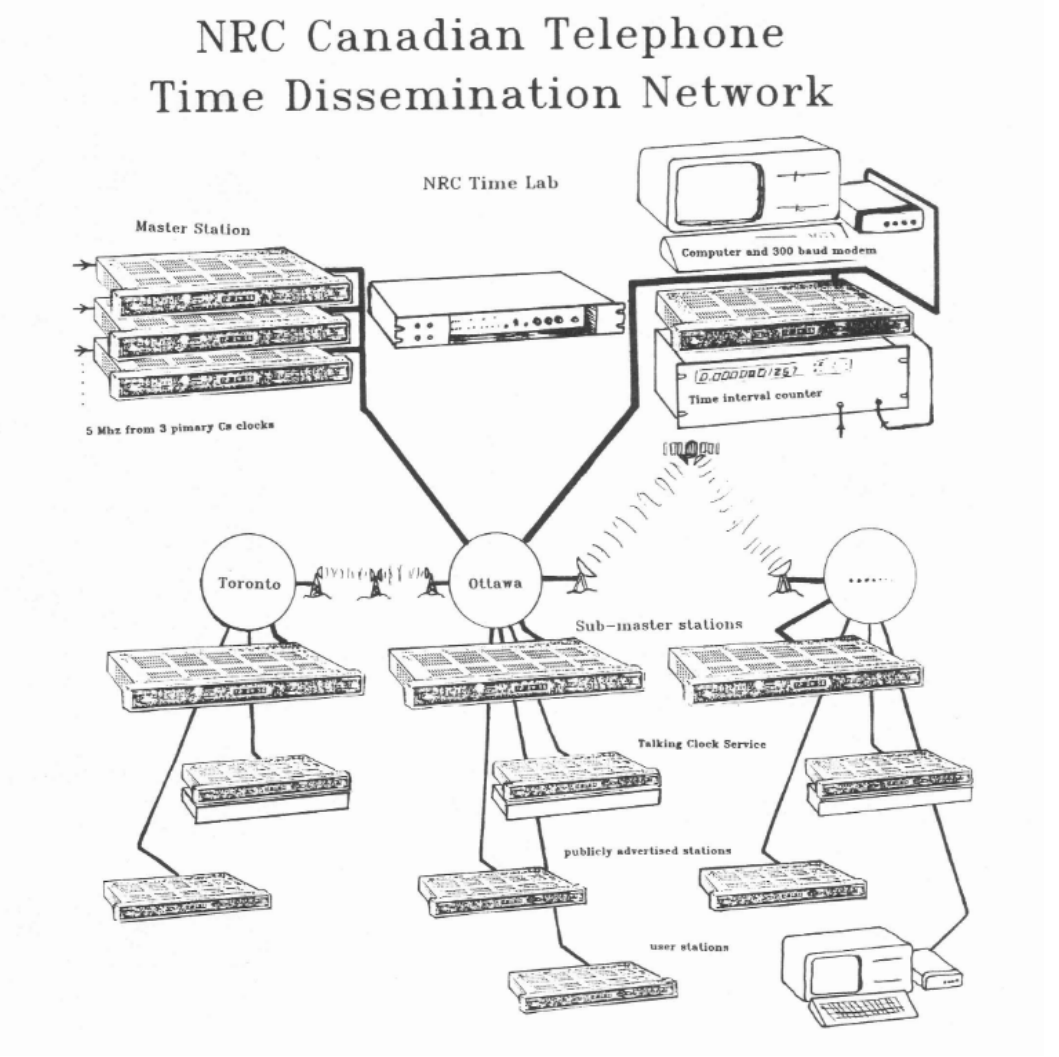
Leitch CSD-5300 - Clock System Driver

- 1985 device supplying time code for slave clocks, broadcast equipment, etc.
- Learned about this via **@themaritimegirl** on YouTube
- The CSD-5300 can call another CSD-5300 via 300 baud modem to set its own time



Leitch CSD-5300

- The National Research Council of Canada still runs CSD-5300s to call for time
  - (613) 745-3900
- The NRC proposed a Telephone Time Dissemination Network in the ‘80s that never came to fruition
- Different than the NIST’s ACTS
  - (303) 494-4774 (CO)
  - (808) 335-4721 (HI)



Leitch CSD-5300

- My CSD-5300 takes calls, **267-417-TIME (8463)**
- Other than the NRC, **@themaritimegirl** and I have the only publicly-accessible CSD-5300s operating
- Finally time for a Telephone Time Dissemination Network?



Telecommunications Devices for the Deaf

- Electronic device for text communication over a telephone line, that is designed for use by persons with hearing or speech difficulties
- Also known as a Teletypewriter (TTY)
- Uses baudot code at either 45.45 or 50 baud (just like RTTY)
- Have screens, printer, full keyboard





Telecommunications Devices for the Deaf

- Baudot-compatible modems do exist!
- Pricey/expensive, somewhat hard to find
- Not reliable, all of mine have issues



TDDs as Terminals

- TDDs are just compact computer terminals!
- Common in many airports
- Supported by Asterisk Softmodem

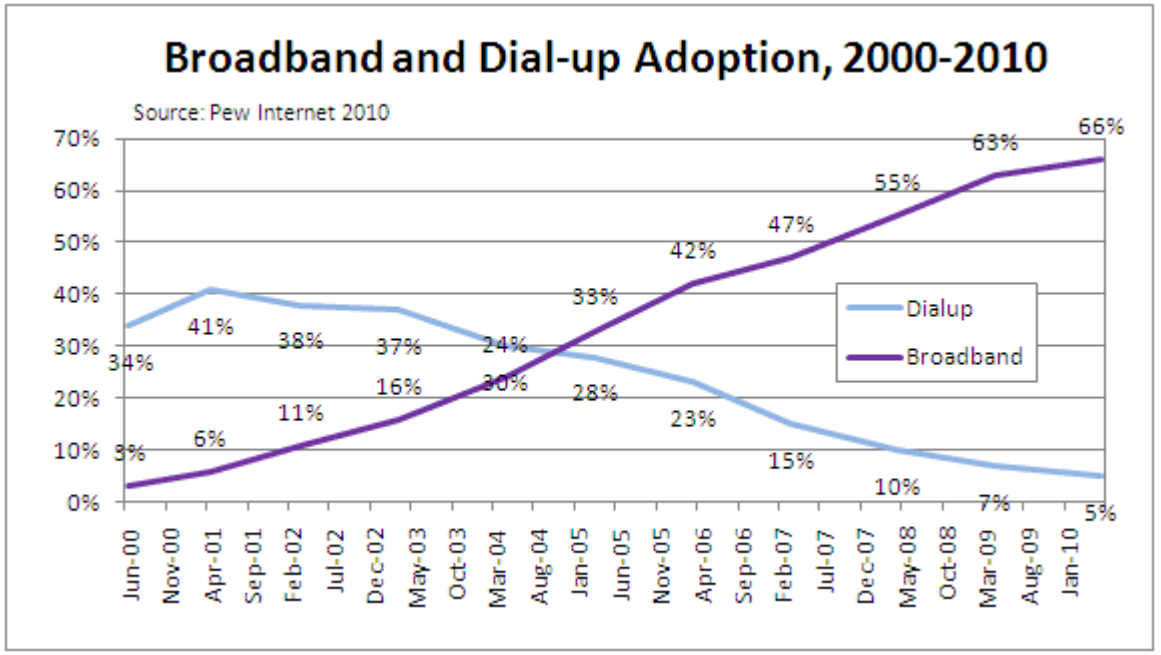


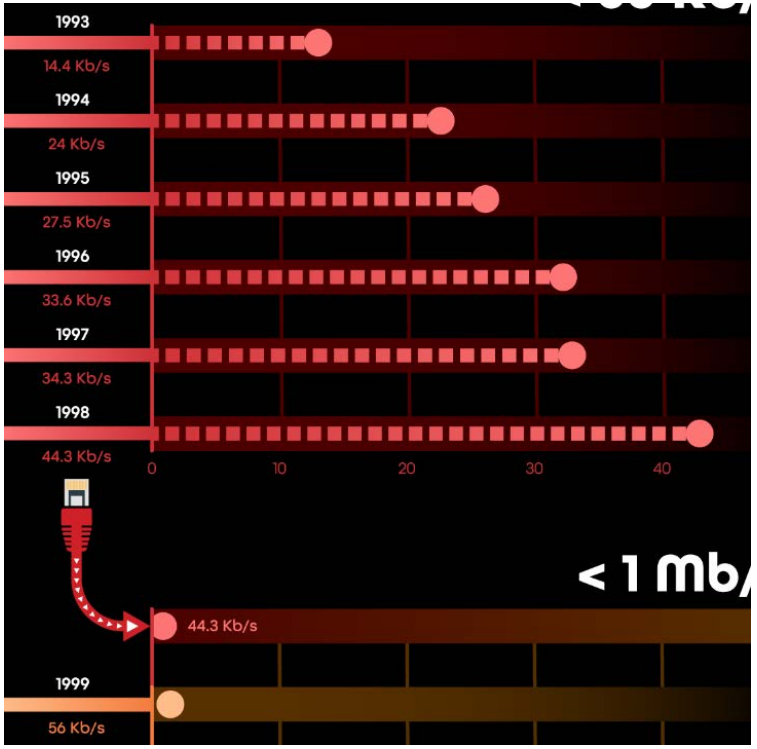
(301) 301-1337



Dial-Up Internet Access

- Modem connection to an ISP, as opposed to a “direct dial” link
- 56k not common until the end of the 90s

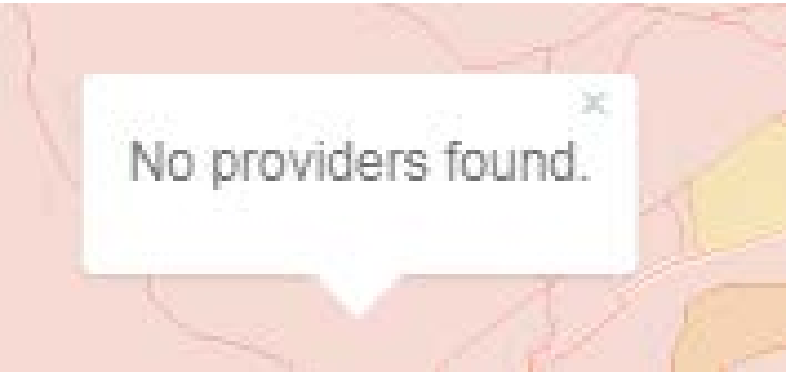




Dial-Up... In the 2020s?

- Still common in areas unserved by broadband
- Sometimes chosen for its cost-effectiveness
- Every dial-up ISP resells the same wholesale provider

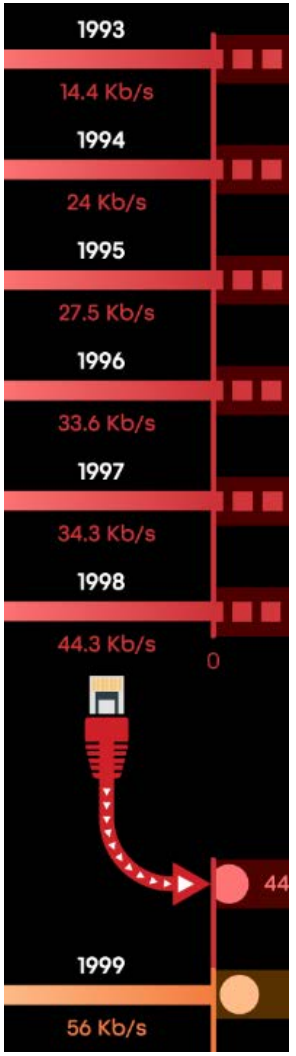
GLOBALPOPS
THE COMMON SENSE CARRIER



Dialup		
Choose	Speed [max]	Price per month
☐	56K v.92	$10

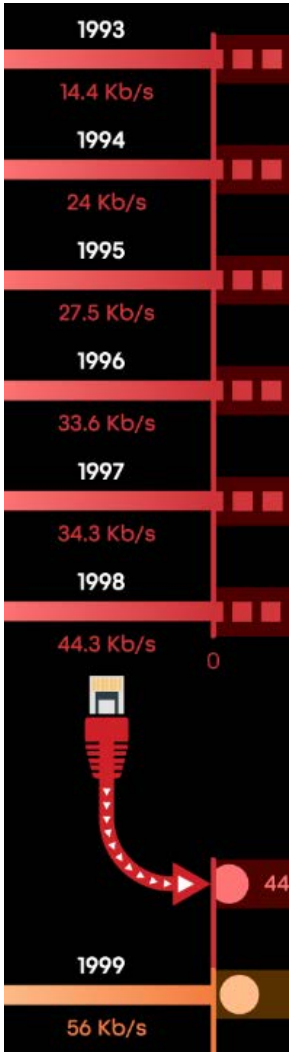
Dial-Up Speeds Today





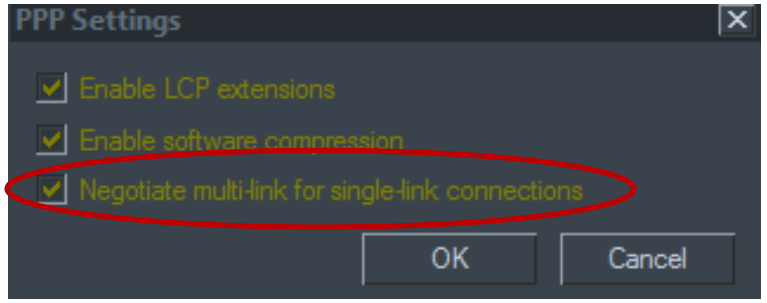
That Darn “33.6k Cliff”

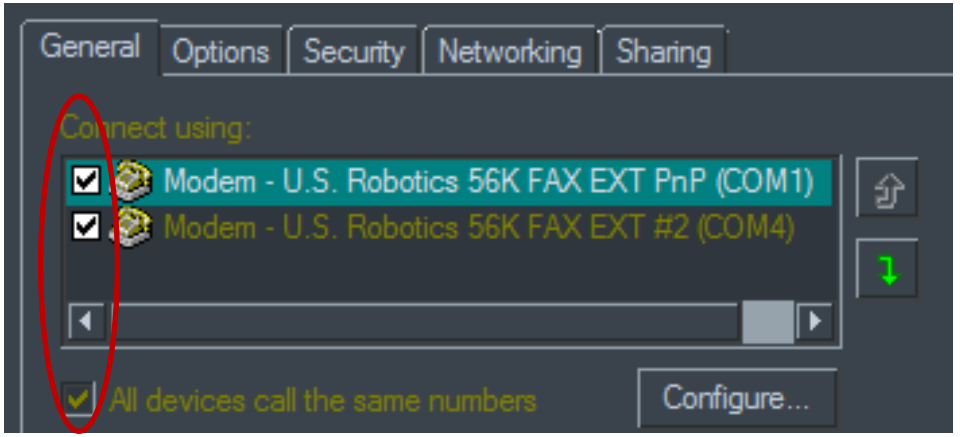


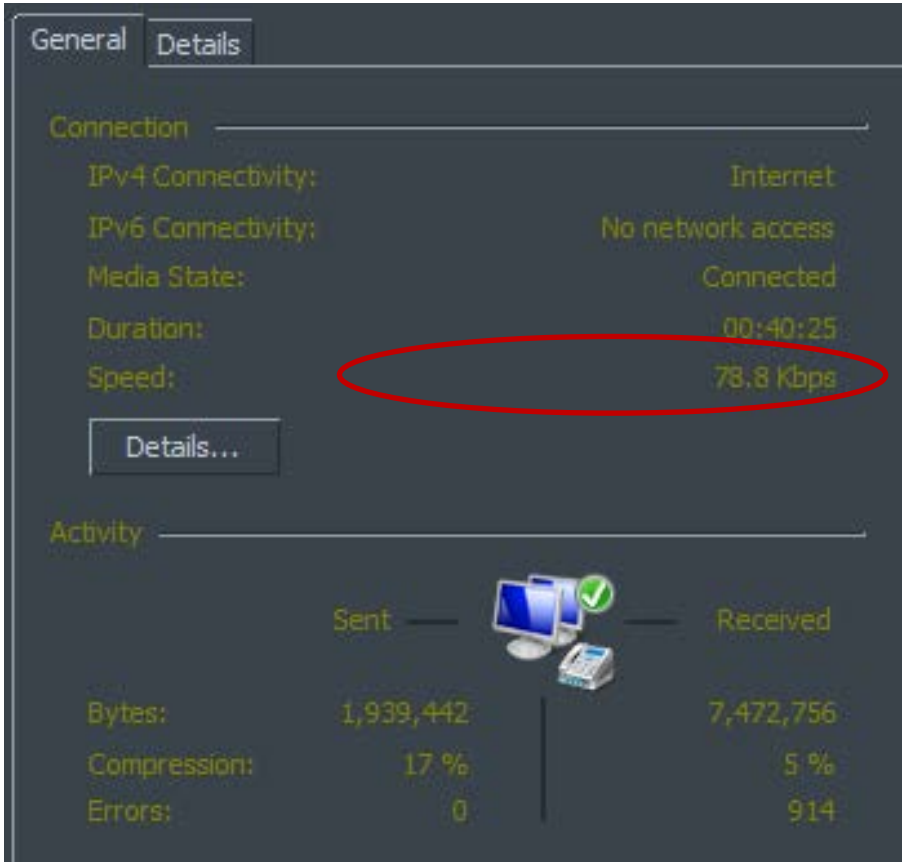


Multilink PPP

- Bundles bandwidth from multiple phone lines together
- Requires support by the ISP to bundle concurrent sessions
- Also called “bonding”







“High-Speed Dial-Up” - A True Story From 2021

- One of my users ordered some “extra” phone lines for multilink
- 8 total POTS line, 7 for dial-up = (up to) ~350 kbps!

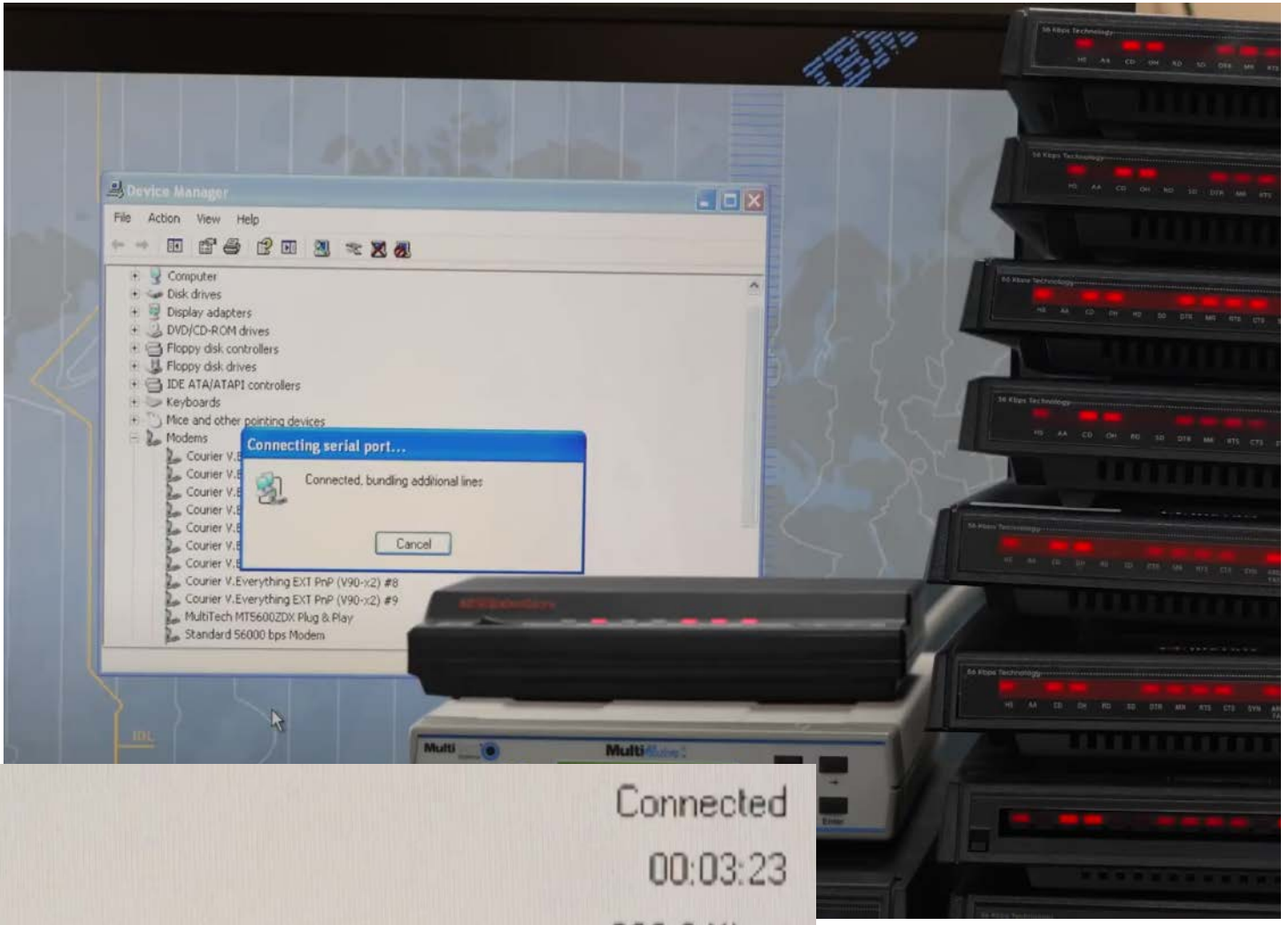
```
online(6) 00:13:21 hours stillActive(0)    v92(22)    48000 0.0.0.0    0
online(6) 00:13:20 hours stillActive(0)    v92(22)    46666 0.0.0.0    0
online(6) 00:13:20 hours stillActive(0)    v92(22)    46666 0.0.0.0    0
```

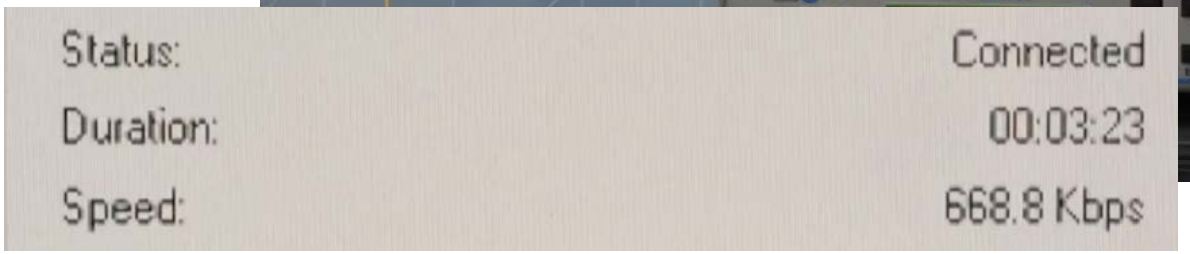
324.5 kbps with seven lines connected!

It says I’m getting 349.1 kbps on the PC connection status

It definitely feels faster- literally like high speed dial up It’s great- amazing what is possible - most everything is much more manageable.

So, how many lines *can* you bundle?





https://www.youtube.com/@theserialport

Future Plans

- Does 56K work over VoIP? Yes, but it is more difficult
  - We can leverage TDM equipment and digital modems
  - There are a number of options!
- Evaluate dial-up as a primary Internet connection



Questions?

Mike

- mike@philtel.org
- Famicoman (Libera IRC)
- https://famicoman.com
- https://github.com/Famicoman

Naveen

- interlinked@philtel.org
- InterLinked (Libera IRC)
- https://interlinked.us
- https://github.com/InterLinked1

